

ColorAptitude — CAP-W10

A digital platform for professional colour competency assessment based on a three-layer cognitive framework, OKLCH-based stimulus generation, and differentiated per-layer validation

Technical Whitepaper — Scientific Disclosure & IP Positioning

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1. Abstract

This document describes a computer-implemented digital platform for professional colour competency assessment, training, and certification. It is designed for the qualification of visual observers engaged in colour-critical work — automotive coatings, textile production, graphic arts, food and cosmetic quality control, conservation — and is positioned as an occupational-qualification instrument in the sense of ASTM E1499-16 rather than as a clinical screening or grading test such as the FM100, Cambridge Colour Test, or Colour Assessment and Diagnosis test.

The platform is organised around a three-layer cognitive framework decomposing professional colour competency into discrimination (detecting that a colour difference exists), attribution (identifying which perceptual dimension has changed), and communication (translating the percept into standardised professional terminology). The layers are measured by dedicated instruments, scored through independent pipelines, and reported as separate values. A 3×3 competency matrix combines the three layers with hue, lightness, and chroma, producing nine resolved cells per session that support targeted training selection, cell-resolved certification scope, and cross-session progress tracking.

Stimulus generation uses OKLCH with per-hue gamut-aware trial construction, ensuring that each trial varies exactly one perceptual dimension without cross-dimensional leakage from clipping. A two-path display calibration architecture accommodates observer-managed deployment (Path A) and ASTM E1499-16 §5.3 compliant qualification (Path B), with unified scoring and certificate-tier annotation. Validation follows a differentiated three-track strategy reflecting the distinct psychometric development stages of the layers. The temporal architecture of assessment, training, and re-test is grounded in the consolidation kinetics of perceptual learning, with platform-level scheduling rules for inter-session intervals, training-to-retest cooldowns, retention probes, and warm-up phases.

The subject matter disclosed in Sections 3.1 through 3.9 comprises the three-layer framework, the 3×3 matrix reporting architecture, the differentiated validation strategy, a five-segment 18-cap Hue Ordering Diagnostic with three red-green opponent segments, the two-path calibration architecture, cardinal-nearest hue classification avoiding Logvinenko circularity, OKLCH gamut-aware trial generation, Kuehni-anchored closed-set attribution scoring, and a temporal architecture for assessment, training, and re-test grounded in perceptual-learning consolidation kinetics. A companion scientific manuscript describing the same subject matter in its academic context has been prepared for submission to Colour Research and Application.

2. Technical Background

This section establishes the minimum context required to read Sections 3.1 through 3.9 without external reference. Readers seeking fuller treatment are referred to the companion scientific manuscript.

2.1 Professional colour competency as a three-part capability

Professional colour judgment in industrial contexts requires three empirically distinguishable capabilities.

Discrimination is the perceptual ability to detect that a colour difference exists, grounded in retinal and early cortical

processing. Attribution is the cognitive-analytical ability to identify which perceptual dimension (hue, lightness, chroma) has changed and in which direction; neuropsychological double-dissociation evidence and pre-verbal infant categorical perception establish this layer as separable from discrimination at the neural level. Communication is the linguistic-professional ability to translate perceived differences into standardised terminology and to reconstruct a percept from a description; cross-cultural colour-naming research and naming stability across the ageing lifespan establish this layer as separable from both lower layers.

Existing clinical instruments (FM100, Cambridge Colour Test, Colour Assessment and Diagnosis test) measure discrimination only. No commercially available instrument known to the author combines all three capabilities in an integrated assessment platform before this publication.

2.2 Colour dimensions and perceptually uniform colour spaces

Three dimensions are inherent in any cylindrical perceptual colour space: hue, lightness, and chroma. HSB and legacy sRGB are not perceptually uniform. CIELAB is approximately uniform but exhibits hue non-uniformities in the blue region. OKLCH (Ottosson, 2020), derived through a CAM16-aware non-linear transformation of Oklab, provides perceptual uniformity sufficient for single-dimension trial generation. OKLCH is used throughout Section 3; OKLCH itself is prior art of Ottosson and is not claimed.

2.3 ASTM E1499-16 as a qualification framework

ASTM E1499-16 specifies observer-qualification requirements across selection, baseline evaluation, training, formal qualification, retraining, and re-qualification. Sections relevant here include §5.3 (instrument compliance), §6.3 (discrimination testing), §6.4 (magnitude scaling), §7.1.1 (warm-up observations), §7.2 (training integration), and §8.1.1 (session limits). The standard is prior art and is not claimed; implementation of its requirements in a specific integrated digital platform combined with the novel methods of Section 3 is the subject of this publication.

2.4 Conventions used in Section 3

Each of the nine innovations follows the same internal structure: Background, Method description, Implementation, and Alternative implementations. Terminology and mathematical conventions follow ISO/CIE standards where available; non-standard terms are defined on first use.

3. Disclosed Innovations

3.1 Three-Layer Competency Framework for Colour Assessment

Background

Existing digital and physical colour-vision instruments measure discrimination only, through ordering or threshold tasks. No existing architecture distinguishes between perceptual discrimination, analytical attribution, and linguistic communication as separable and separately scored components of professional colour competency. The absence of this distinction prevents differentiation between observer profiles that behave differently in practice: the observer who detects but cannot describe, the observer who describes but cannot judge magnitude, and the observer who does both reliably.

The Japanese Colour Aptitude Test (Hirschler, Gay & Ferreira de Oliveira, 2002) includes a Colour-Attribute Discrimination Test as a physical-artefact precedent for attribution measurement, but it is not integrated with discrimination or communication measurement, is not digital, and does not produce dimension-resolved or layer-resolved scoring. Neuropsychological evidence for the separation of attribution from naming (Siuda-Krzywicka et al., 2019; Skelton et al., 2017; Hardy et al., 2005) establishes the empirical legitimacy of the three-layer distinction but has not been operationalised as an assessment architecture.

Method description

The method measures three capabilities as separate competencies, each with its own stimulus structure, task demand, scoring pipeline, and reporting output.

The discrimination layer measures detection of a colour difference. Responses are spatial arrangements, forced-choice judgements, or thresholds from adaptive procedures. No linguistic or categorical response is required.

The attribution layer measures identification of the changed dimension between two stimuli that differ on exactly one dimension. Stimulus pairs are constructed such that hue, lightness, or chroma varies in isolation. Responses identify the changed dimension, optionally with direction and magnitude.

The communication layer measures translation of a percept into standardised terminology and reconstruction of a percept from a description. Tasks proceed in colour-to-description direction (closed-set descriptor selection) or description-to-colour direction (closed-set candidate selection). Vocabulary is professionally standardised, addressing the methodological concern of Logvinenko (2012) regarding free-choice naming.

Each layer is scored through an independent pipeline that does not aggregate across layers. The session report presents three layer-level scores visible to administrator, observer, and downstream systems. Aggregation to a single index may be performed for summary purposes but does not replace the layer-resolved report.

Implementation

The reference implementation is the ColorAptitude platform. The discrimination layer is implemented through the Hue Ordering Diagnostic (§3.4), a Paired-Comparison Instrument for lightness and chroma thresholds, and supporting procedures. The attribution layer is implemented through Dimension Identification and Magnitude Attribution Instruments using cardinal-nearest classification (§3.6, §3.8). The communication layer is implemented through a Colour Naming Instrument anchored to the Kuehni cardinal-hue partition (§3.8) and a Descriptor Selection Instrument with direction-and-magnitude descriptors. Session output is the 3×3 matrix of §3.2.

Layer-differentiated training selection follows from the layer-resolved report: an observer scoring high on discrimination but low on attribution receives analytical training; high on attribution but low on communication receives vocabulary training. Single-score instruments cannot support such targeting because the differentiating information is not preserved.

Alternative implementations

The architecture does not require the specific instruments of the reference implementation. Discrimination may be measured by any ordering, comparison, or threshold paradigm. Attribution may be measured by any forced-choice procedure isolating single-dimension variation. Communication may be measured through any closed-set linguistic protocol, in any natural language, with any standardised vocabulary coherent with the curriculum. The stimulus-generation infrastructure may use any perceptually uniform colour space supporting per-hue gamut-aware construction; OKLCH is suitable but not unique.

The methodological commitment is the architectural separation of the three capabilities into independently measured, scored, and reported components. Any implementation preserving this separation instantiates the disclosed method.

3.2 Three-by-Three Competency Matrix as Reporting Architecture

Background

Existing instruments report a single global score (FM100 Total Error Score) or a few confusion-axis indices (Cambridge protan/deutan/tritan). Neither distinguishes which perceptual dimension a deficit concentrates in nor which cognitive layer it operates at. An observer with strong hue but weak lightness discrimination, or with strong

chroma attribution but weak chroma communication, cannot be identified from a global score and cannot be assigned targeted training, cell-specific certification, or dimension-layer progress tracking.

Method description

The method produces a 3×3 reporting structure with three rows (discrimination, attribution, communication) and three columns (hue, lightness, chroma). Each cell contains a score derived from instrument responses assigned to that layer-dimension combination. The matrix is produced at session end and preserved through all downstream processing. Aggregation to row-level layer scores or to a single matrix-level index may be performed as supplementary summary but does not replace the cell-resolved report at any pipeline stage.

Certification decisions are made at cell level. A certificate may state that an observer meets criteria for hue discrimination and hue attribution but not for chroma communication. This cell-resolved scope supports professional contexts requiring specific competency combinations: textile colourist roles may require all nine cells; graphic-design roles only the hue row; colour-inspection roles only the discrimination row. Targeted training selection follows from the cell-resolved gap analysis. Cross-session progress tracking is performed per cell, revealing whether training has affected the targeted competency or shifted other cells as a side effect.

Implementation

The reference implementation produces per-cell scores using the PCC-D, PCC-A, and PCC-C metrics described in the companion Paper 3. PCC-D for hue derives from segment-normalised TES on the Hue Ordering Diagnostic; PCC-D for lightness and chroma from reference-normalised thresholds. PCC-A cells derive from chance-corrected accuracy on Dimension Identification and Magnitude Attribution Instruments. PCC-C cells derive from closed-set sector accuracy on the Colour Naming Instrument and from descriptor accuracy on the Descriptor Selection Instrument. Each cell is expressed as a score in [0, 1] with a tier label (unvalidated / provisional / established) reflecting normative-data state.

Alternative implementations

The 3×3 architecture is independent of specific scoring formulas. Any method producing per-cell numerical values from instrument responses and preserving them through reporting instantiates the architecture. The dimensional structure corresponds to the three principal perceptual dimensions of a cylindrical colour space and is not tied to any specific representation. Extensions to additional rows (a fourth layer for colour-constancy competency) or columns (a fourth dimension for gloss or texture) are compatible provided the per-cell reporting is preserved.

3.3 Differentiated Three-Track Validation Strategy

Background

Conventional psychometric validation is treated as a single activity assuming all measured capabilities sit at comparable development stages. The three-layer framework violates these assumptions. Discrimination has a century-old tradition with extensive normative data (Kinnear & Sahraie, 2002; Verriest et al., 1982). Attribution has no established normative population because it has not been systematically measured at population scale; validation depends on known-groups comparisons against independently defined expert criteria. Communication has no pre-existing population of trained observers at the scale required for norm-referenced validation; that population is in part what the methodology is designed to create. A monolithic strategy cannot serve all three layers.

Method description

The method uses three parallel tracks, one per layer, each matched to that layer's psychometric development stage.

Track 1 — Norm-referenced validation for discrimination.

A reference sample of $N \geq 200$ industrial practitioners, stratified by age decade and sector, is administered the discrimination instruments under a two-session protocol with a 7-to-21-day inter-session interval. The protocol produces age- and sector-stratified normative distributions, test-retest reliability estimates, and convergent-validity evidence against the Farnsworth-Munsell 100 Hue Test for a subsample. The lower bound of seven days is informed by the consolidation kinetics established by Karni and Sagi (1993): a slow between-sessions learning component completes consolidation within 22-28 hours of an initial exposure and remains stable for at least 22-32 months thereafter. An interval of seven days or more therefore falls well outside the consolidation window while remaining short enough to make structured deliberate practice between sessions impractical. Initial norms are expected within ~18 months.

Track 2 — Known-groups validation for attribution.

A pilot study with three groups (trained-current, trained-non-current, untrained) of $N \geq 30$ each, matched on age, sex, sector, and visual acuity, is administered the attribution instruments. Expert-status requires ≥ 40 hours of formal colour training covering all three dimensions and current employment in a colour-critical role with documented attribution-level reporting. Between-group effect size $d \geq 0.8$ is the pre-specified threshold for demonstrating attribution as distinct from discrimination. Pilot result expected within ~12 months.

Track 3 — Longitudinal construct-validity for communication.

Validation accompanies curriculum deployment because the curriculum-trained population does not yet exist at norm-referenced scale. Four convergent evidence lines accumulate: inter-rater agreement on shared stimuli, within-observer stability over repeated administration, predictive validity for industrial reporting quality, and cross-cultural replication across language contexts. Tier-labelled reporting accompanies the evidence: cell scores are labelled "preliminary" during early deployment, "emerging" once inter-rater and within-observer evidence accumulates, and "established" once predictive-validity has converged in at least two industrial contexts.

The three tracks proceed at different paces and produce different evidence types. Tier-labelled cell scores allow all three layers to contribute to the competency profile at each validation stage without overreaching the available evidence for any layer.

Implementation

The reference implementation specifies the tracks as parallel, separately managed empirical programmes. Track 1 is an ongoing normative-reference sampling operation; Track 2 a discrete pilot with cross-sector replication options; Track 3 a continuous evidence-accumulation programme linked to curriculum deployment. Session-level reporting includes the current validation tier per cell, stored as metadata and updated as evidence accumulates.

Alternative implementations

The differentiation is driven by psychometric development stages, not by specific sample sizes, effect-size thresholds, or time horizons. Alternative parameters and tier-labelling conventions are compatible provided the core commitment — validation method matched to development stage — is preserved. The method generalises to other multi-layer assessment frameworks where layers sit at different psychometric stages, including medical education and skills-based qualification.

3.4 Five-Segment 18-Cap Hue Ordering Diagnostic with Three Red-Green Opponent Segments

Background

The Farnsworth-Munsell 100 Hue Test organises 85 movable caps across four trays whose boundaries lie at the four unique hues. Two of these segments cross red-green opponent transitions (red-yellow, green-blue); two do not. Red-green deficiency produces ordering errors concentrated in the opponent-crossing segments, but with only two

such segments available the diagnostic signal is bounded by the number of independent observations of the red-green axis.

A diagnostic providing additional independent observations of the red-green axis, while remaining comparable in duration and procedure to the FM100, would increase diagnostic power without compromising the established paradigm. Fairchild (2018) argued that discrimination tasks align more closely with the five Munsell principal hues (red, yellow, green, blue, purple) than with the four unique hues, because principal-hue boundaries lie at established colour-order-system landmarks rather than at observer-dependent unique-hue loci.

Method description

The method organises the hue circle into five segments at the five Munsell principal hues after OKLCH conversion: red-to-yellow, yellow-to-green, green-to-blue, blue-to-purple, purple-to-red. Three segments cross red-green opponent transitions (green-to-yellow, blue-to-green, purple-to-red); two do not. The three opponent-crossing segments provide three independent observations of the red-green axis per administration. A red-green deficiency showing systematically across all three opponent segments — relative to the two non-opponent segments — is identifiable with higher confidence than the same deficiency mapped onto a four-segment structure. The diagnostic gain is the number of independent observations, not segment-arrangement symmetry.

Each segment contains 18 caps: 17 movable plus one fixed boundary anchor. Total: 90 caps, 85 movable. The 85-movable count preserves comparability with the FM100 while reorganising into the five-segment structure. Session duration is approximately 6% longer than the FM100 for a comparable population.

Error scoring follows Farnsworth (1957) per movable cap, summed across segments. The worst-case TES per 18-cap segment is analytically derived as $2 \cdot (\lfloor 324/2 \rfloor - 1) - 2 \cdot 17 = 288$. Worst-case across all five segments is 1440. The ratio to FM100 worst-case (1584) is $1440/1584 \approx 0.91$, supporting linear rescaling of FM100 normative data as a provisional reference pending instrument-native norms.

Cap visual angle is 1° - 1.5° at the observer's eye, with viewing distance 50-60 cm. The specification targets equivalence with the physical FM100 (caps $\sim 13 \times 15$ mm subtending $\sim 1.5^\circ$ at 50 cm). The upper bound is informed by evidence that hue-shift and saturation effects become substantial as field size grows beyond the CIE 1931 2° standard observer (Hsieh & Chen, 2010). Evidence that inter-observer variability increases substantially below 2° (Wyszecki & Stiles, 1982; Li, Hanselaer, & Smet, 2023) derives from binocular matching paradigms and does not transfer directly to the sorting paradigm of the Hue Ordering Diagnostic.

Implementation

The reference implementation generates the 90 cap stimuli through OKLCH-based per-hue gamut-aware construction (§3.7). Cap hue angles are computed from the five-segment Munsell anchors, interpolated linearly within each segment. Cap lightness and chroma are held constant at the §3.7 reference values to isolate hue. Administration proceeds segment by segment with randomised segment order. Per-segment and total TES are reported; per-segment scoring feeds the PCC-D hue score (Paper 3). Cap dimensions in pixels are computed from detected DPI and observer-supplied viewing distance; sessions on displays where the target cap size is not achievable with all 18 caps simultaneously visible are flagged.

Alternative implementations

The five-segment structure is the methodologically committed feature. Alternative cap counts per segment (e.g., 16 or 20) are compatible provided the five-segment structure with three red-green opponent segments is preserved. The Munsell-principal-hue segment placement is the disclosed feature; alternative landmark systems (e.g., NCS elementary hues) produce different segmentations and are distinct methods. The TES value of 288 per 18-cap

segment is a property of the cap count specifically; alternative counts produce different worst-case values via the same formula. The 1440-to-1584 FM100 ratio is a property of the 5×288 configuration.

3.5 Two-Path Display Calibration Architecture with ISO 3664:2009 P2 Compliance Logging

Background

Digital colour-assessment instruments on consumer displays face a methodological dilemma. Hardware calibration with spectrophotometers or colorimeters supports formal ASTM E1499-16 §5.3 qualification but is unavailable to most observers. Software-only calibration produces acceptable approximations for training but does not meet §5.3. An instrument supporting only hardware calibration is restricted to laboratory and enterprise deployment; one supporting only software cannot produce qualifying scores. A two-path architecture making both available, with session-level recording of the active path, accommodates the deployment range from formative self-training through formal qualification.

Method description

The method implements two parallel calibration paths producing identically structured scoring output. The active path is recorded as session metadata and preserved through certification.

Path A — Software-indicative calibration.

The observer completes a seven-step calibration wizard adjusting display parameters via reference stimuli and observer responses. The wizard records adjusted gamma, estimated white-point CCT, and estimated black-level emission. Output is labelled "indicative" in all reports and is suitable for within-observer tracking, formative feedback, training selection, and cross-session progress assessment. It is explicitly not suitable for formal qualification under ASTM E1499-16 §5.3.

Path B — Hardware-compliant calibration.

A Spyder/i1-class colorimeter measures display white point, gamma, and primary chromaticities to §5.3 precision. The colorimeter's ambient sensor measures ambient illuminance and CCT at the observer position. Output is labelled "hardware-compliant" and is the basis for formal qualification.

Path B additionally implements ISO 3664:2009 P2 viewing-condition compliance logging. The standard specifies an ambient-to-display luminance-ratio bound of 1/4 (mandatory) or 1/8 (recommended) for softproof. Sessions with ratios above the recommended bound are flagged with a compliance note; sessions above the mandatory bound as non-compliant. The ratio is computed in real time from white-point luminance and ambient illuminance at screen-height reference plane.

Path B additionally logs mixed-chromatic-adaptation risk. When ambient CCT differs substantially from display white-point CCT, the visual system adapts partially to each, producing approximately 60/40 display/ambient adaptation under canonical mixed conditions (Katoh et al., 1998; Peng et al., 2021). Sessions with substantial CCT divergence are flagged with a risk note that propagates into the session report as an interpretive caveat for cells near adaptation-sensitive regions.

Certification tiers reflect the active path. A training-tier certificate may be issued under Path A; a qualification-tier certificate requires Path B and ISO 3664:2009 P2 compliance. The two-path distinction is visible in the certification record so that downstream consumers (employers, regulators, audit systems) can verify the issuance basis.

Implementation

Path A is supported through a web-based seven-step wizard running in any modern browser without additional software. Path B integrates with Spyder- and i1-class colorimeters via vendor SDKs. The active path is recorded as a session-metadata field cryptographically bound to the score record at session-close, preventing retroactive

relabelling. ISO 3664:2009 P2 compliance and mixed-adaptation risk are computed in real time and displayed during session setup with suggested mitigations.

Alternative implementations

The architecture is committed to the distinction between software-indicative and hardware-compliant calibration as separable, operationally parallel modes. The seven-step Path A structure, specific colorimeter integrations, and specific ratio thresholds are implementation details. Alternative wizard structures, alternative colorimeter integrations, and future ISO 3664 revisions (including ISO 3664:2025 where adopted) are compatible provided the two-path separation and metadata recording are preserved. The architecture generalises to any digital psychometric instrument whose scoring validity depends on measurable deployment conditions.

3.6 Cardinal-Nearest Hue Classification for Attribution Scoring

Background

Hue-attribution scoring requires a reference partition of the hue circle. Two conventional approaches present methodological problems for qualification testing.

Free-naming, with responses scored against a lexicon of acceptable names, suffers the circularity formalised by Logvinenko (2012): the observer's naming becomes both measurement and reference, so an observer systematically biased relative to population consensus is undetectable because the reference moves with the bias.

Observer-derived unique-hue calibration, using the observer's own unique-red, unique-yellow, unique-green, and unique-blue selections as reference, avoids free-naming but reintroduces the circularity at the calibration stage: an observer with shifted unique-hue judgements is scored against their own shifted reference and the shift is invisible.

A method using a fixed reference set derived from empirical population data, independent of the observer under test, avoids both circularities while preserving diagnostic information.

Method description

The method uses the mean-observer cardinal-hue angles reported by Kuehni, Hinks, and Shamey (2008) as the fixed reference partition. Kuehni et al. (2008) collected unique-hue selections from a large trichromatic sample and computed population-mean angles for unique red, yellow, green, and blue. These define a fixed four-sector partition with boundaries at the angular midpoints between adjacent cardinals.

For each attribution trial, the OKLCH stimulus hue angle is classified to the nearest Kuehni cardinal by angular distance. The observer selects from a closed set of named hue regions corresponding to the four cardinal sectors. A matching response is scored correct; an adjacent-sector response is scored partially correct (boundary tolerance for near-boundary stimuli below a threshold); a non-adjacent response is scored incorrect.

The classification is cardinal-nearest at both stimulus and response side: stimuli are classified to the nearest Kuehni cardinal; responses select from a closed set of sector names corresponding to the same regions. The observer is never asked to free-name and never to select a unique hue as calibration anchor; both circularity-introducing moves are avoided. Population-level bias becomes a measurable property of the observer rather than an invisible offset absorbed into self-referential scoring.

Implementation

The reference implementation uses the Kuehni et al. (2008) angles as the fixed cardinal set. Stimulus hue angles are converted from OKLCH to Munsell-compatible hue coordinates for comparison and sector membership is computed analytically. The closed response set uses professionally standardised names (red, yellow, green, blue) matching the four cardinal regions. Boundary tolerance applies to stimuli within 10% of sector width from a sector boundary;

adjacent-sector responses within this band score as partially correct (half weight). The PCC-C communication score and the hue-attribution score both consume the classification output (Paper 3).

Alternative implementations

The method is committed to a fixed population-derived observer-independent partition. The Kuehni et al. (2008) choice is the reference implementation's; alternative population datasets (Xiao et al., 2011; Shamey et al., 2010) produce slightly different cardinals and are compatible provided observer-independence is preserved. Alternative partition granularities (eight sectors using principal plus intermediate cardinals) are compatible; chance baselines and required sample sizes scale accordingly. The method extends to other attribution dimensions where population-derived references are available, including lightness, chroma, and composite perceptual regions.

3.7 OKLCH Gamut-Aware Per-Hue Trial Generation with Binary-Search Safe Chroma

Background

Single-dimension stimulus variation in OKLCH is arithmetically straightforward — dimensions are by construction separately addressable. But the reproducible gamut of a display does not fill the OKLCH cylinder uniformly. Some hue-lightness combinations admit chroma near full saturation; others admit only reduced chroma before a coordinate falls outside sRGB or P3 and is clipped. Clipping introduces cross-dimensional variation: a trial intended to vary hue at constant chroma may in fact vary both because different hue angles have different gamut limits at the selected lightness. The intended dimensional isolation is compromised and observer-response interpretation becomes ambiguous.

Method description

The method computes a per-hue maximum reproducible chroma for a given lightness across the full hue-angle set in a trial set. Per-hue maxima are computed by binary search: the OKLCH coordinate is converted to sRGB (or target gamut) and tested for membership at successively larger chroma values until the largest in-gamut chroma is identified to a pre-specified precision.

The applied chroma for the trial set is then 95% of the minimum in-gamut chroma across all hue angles in the set. The 5% safety margin accommodates floating-point rounding in the OKLCH-to-sRGB pipeline, sub-pixel rendering variation at the display surface, and inter-display gamut differences. The minimum-across-hues selection guarantees that every hue angle can be rendered at the applied chroma without clipping.

Hue-variation trials use the applied chroma uniformly. Chroma-variation trials use a graduated scale bounded by the per-hue maximum. Lightness-variation trials use the applied chroma computed for the full lightness range under test (the minimum-across-lightnesses extension).

The reference lightness for the Hue Ordering Diagnostic and related discrimination instruments is $L = 0.70$ in OKLCH. The choice balances three considerations: in-gamut chroma is relatively large across the hue circle at $L = 0.70$, maximising discriminable headroom; $L = 0.70$ approximates Munsell Value 7, a commonly used reference lightness in industrial colour evaluation; and OKLCH perceptual uniformity is most reliable in the middle L range where calibration data is densest. The value is provisional pending empirical work on alternative reference lightnesses.

Implementation

Conversion uses the Ottosson (2020) reference OKLCH implementation. Binary-search precision is 10^{-4} in chroma, converging in ~ 14 iterations per hue angle. The 5% margin is applied as post-conversion reduction. For the Hue Ordering Diagnostic, per-hue maxima are computed across the 90 hue angles at $L = 0.70$. For the Paired-Comparison Instrument, computation is at the trial-anchor angles at $L = 0.70$. Lightness-variation trials use the minimum-across-

lightnesses reduction. Rendered trials are verified at render time by reverse OKLCH conversion; verification failures trigger session abort and recalibration prompt.

Alternative implementations

The method is committed to keeping chroma values within the reproducible gamut across all stimuli in a trial set without clipping. The 95% safety margin is an implementation choice; 90% (more conservative) or 98% (tighter utilisation) are compatible. The $L = 0.70$ reference is provisional; alternative lightnesses are compatible provided per-hue minimum-chroma computation is performed at the chosen lightness. Alternative perceptually uniform cylindrical spaces (CIECAM02-UCS, CAM16-UCS, CIELCh) and alternative display-gamut targets (Display P3, Rec. 2020) are compatible provided the conversion pipeline is appropriate. The method extends to any application requiring single-dimension variation in a bounded multidimensional space.

3.8 Kuehni-Anchored Closed-Set Attribution Scoring

Background

Attribution and communication scoring requires correctness determination. For dimensional identification (did hue, lightness, or chroma change?) the criterion is straightforward: stimulus construction defines ground truth. For hue-region identification a reference partition is required, and the choice of partition is methodologically consequential (see §3.6). Closed-set scoring — observer selects from a pre-specified list rather than producing free responses — eliminates interpretive ambiguity at scoring and supports chance-corrected accuracy with known baselines. When the closed set is anchored to population-derived reference data, scoring is methodologically clean (no Logvinenko circularity) and psychometrically defensible (chance baselines known, inter-response distances interpretable).

Method description

The method uses closed response sets for all attribution and communication tasks, anchored to empirical population data where it defines the response options, with chance-corrected scoring accounting for random-response baselines.

For hue-region identification, the closed set comprises four sector names corresponding to the Kuehni cardinal angles (red, yellow, green, blue). Chance baseline is $1/4$. Boundary tolerance (§3.6) handles near-boundary stimuli.

For dimension identification (hue/lightness/chroma), the closed set has three options. Chance baseline is $1/3$. Chance-corrected accuracy is $\max(0, (\text{accuracy} - 1/3) / (1 - 1/3))$, producing a score in $[0, 1]$.

For joint dimension-and-magnitude attribution, three dimensions cross three magnitudes (slight/noticeable/strong), producing nine joint options. Chance baseline is $1/9$. Joint chance-corrected accuracy is $\max(0, (\text{joint_accuracy} - 1/9) / (1 - 1/9))$.

For lightness and chroma descriptor selection, direction (lighter/darker, more/less saturated) crosses magnitude (slight/noticeable/strong), producing six options. Direction and magnitude are scored separately and combined with 0.6/0.4 weighting reflecting that an observer who cannot reliably identify direction cannot usefully estimate magnitude.

The closed-set architecture applies uniformly across attribution and communication, producing scoring outputs directly comparable across task types within each layer and across layers within the matrix. Chance-corrected accuracies in $[0, 1]$ feed PCC-A and PCC-C layer scores (Paper 3).

Implementation

The reference implementation constructs Kuehni-anchored sectors from the mean-observer cardinal angles (Kuehni, Hinks, & Shamey, 2008), computed in Munsell-compatible angle space and converted to OKLCH. The four-sector

partition is stored as a fixed reference table used across all sessions and observers. Response collection presents the closed set as a selection interface (radio buttons or equivalent), constraining one response per trial. Boundary-tolerance scoring is computed at scoring time from recorded stimulus angle and response; boundary regions are not communicated to the observer to prevent strategic adjustment. Chance-corrected scoring is applied at per-dimension and per-cell level before layer aggregation.

Alternative implementations

The architecture is committed to fixed-option selection replacing free-response measurement, with chance-corrected accuracy computed from known baselines. The Kuehni et al. (2008) anchor choice is the reference implementation's; alternative population datasets are compatible. Alternative partition granularities are compatible; chance baselines and scoring formulas scale accordingly. The 2:1 Magnitude-Attribution-over-Dimension-Identification weighting in PCC-A and the 0.6/0.4 direction-over-magnitude weighting in PCC-C are provisional. The method extends to other assessment contexts where response measurement can be replaced by fixed-option selection against empirically derived references.

3.9 Temporal Architecture for Assessment, Training, and Re-Test

Background

Digital platforms integrating measurement, training, and re-test must determine timing relationships between these activities. Three temporal questions arise: how long must the inter-session interval be to avoid contamination by ongoing perceptual learning when an assessment is repeated; how long must the cooldown be after training to ensure training-induced gain has stabilised before re-test; at what interval should a retention probe occur to measure stable post-consolidation capability.

The answers depend on the temporal dynamics of perceptual learning. Karni and Sagi (1993) established three findings that constrain the temporal architecture of any platform operating on perceptual capabilities. First, perceptual learning has two temporally distinct components: a fast within-session phase saturating rapidly during the first practice blocks of the first session and transferring across eyes, and a slow between-sessions phase that is orientation-specific, monocular, and accounts for substantial day-to-day improvements. Second, the slow component is not measurable immediately: a latent phase of approximately eight hours follows practice during which probe testing detects no improvement, with consolidation completing within 22-28 hours. Third, once consolidated, the gain is retained over at least 22-32 months without further training, with almost no forgetting.

Existing instruments do not embody this structure. Clinical tests treat each administration as temporally independent. Adjacent-domain training platforms typically schedule sessions and re-tests by clinical convention or commercial preference rather than by appeal to consolidation theory. A platform explicitly grounding inter-session interval, training-to-retest cooldown, retention-probe interval, and within-session protocol structure in the consolidation kinetics of perceptual learning is, to the author's knowledge, not present in the prior art.

Method description

The method specifies the temporal architecture as four scheduling rules, each grounded in the Karni-Sagi consolidation model.

Rule 1 — Inter-session interval.

The interval between successive assessment sessions for test-retest reliability estimation is bounded below by a minimum placing the second session beyond the slow consolidation window, and above by a maximum within which physiological drift of colour-vision status is unlikely. Seven to twenty-one days satisfies both bounds: the lower bound is approximately seven times the 22-28 hour consolidation window; the upper bound is short enough that age-related or pathological drift is improbable.

Rule 2 — Training-to-retest cooldown.

The cooldown between training-programme completion and a subsequent assessment intended to measure post-training capability is set at thirty days minimum. Thirty days is a conservative multiple of the consolidation window and ensures that perceptual gain attributable to training has fully consolidated, that within-training fluctuations have settled, and that re-test measures the stable post-consolidation state rather than a transient peak or trough of the training trajectory.

Rule 3 — Retention-probe interval.

A retention probe is a short assessment evaluating persistence of training-induced gains, scheduled at approximately fourteen days after training completion. Fourteen days falls well outside the consolidation window so the measured performance represents stable post-consolidation capability, while remaining short enough to detect hypothetical decay before it would substantially affect downstream performance. Alternative intervals at multiples of seven days (7, 14, 21, 30, 60, 90, 180) are compatible; the configured interval is recorded as session metadata.

Rule 4 — Within-session warm-up.

To mitigate contamination by the fast within-session learning component, the scored portion is preceded by an unscored warm-up of practice trials and orienting examples, such that any rapid saturation occurs during warm-up and not during scoring. The warm-up requirement aligns with ASTM E1499-16 §7.1.1 and is grounded here additionally in the consolidation model. The scored portion is bounded in duration to a length over which the slow component is not expected to manifest within a single session — under Karni and Sagi's findings any duration shorter than ~8 hours from session start, easily satisfied by typical sessions of 20-60 minutes.

The four rules compose into an integrated architecture. An observer scheduled for normative-reference assessment under Track 1 of §3.3 completes two sessions separated by 7-21 days (rule 1), each with unscored warm-up before scoring (rule 4). An observer subsequently undertaking training receives re-test eligibility 30 days after training completion (rule 2). An observer in a longitudinal training programme receives retention probes at the configured interval, typically 14 days after each training-block completion (rule 3). The architecture binds the assessment, training, and re-test components into a temporally coherent whole rather than treating them as independent operations.

Implementation

The reference implementation enforces the four rules through platform-level scheduling logic consulting session-completion and training-completion timestamps before permitting subsequent sessions. The seven-day inter-session minimum is a hard constraint: a participant attempting an early second session is shown the earliest permissible date and cannot begin before that date. The twenty-one-day maximum is a soft warning: exceeding it produces a notice that the reliability estimate may include drift contributions, with elapsed-interval recorded as metadata.

The thirty-day training-to-retest cooldown is a hard constraint at the certification pipeline level: an early re-test session produces a score record but the certification engine refuses to issue a post-training certification tier until the period has elapsed and a confirmation re-test has been completed. The participant retains the right to take the re-test for self-monitoring.

The fourteen-day retention probe is a scheduled assessment automatically queued at training-programme completion, with notification at the scheduled time and administration within ± 3 days of the target. Probe results are stored separately from primary results and reported with explicit retention-interval annotation.

The unscored warm-up is the first three to five trials of every session under all instruments, using clearly above-threshold magnitudes (discrimination) or clearly recognisable distinctions (attribution and communication). The

transition to scored phase is silent: the participant is not signalled when scoring begins, preventing strategic effort adjustment.

Alternative implementations

The architecture is committed to grounding scheduling intervals in the consolidation kinetics of perceptual learning. The specific intervals — seven-day minimum, twenty-one-day maximum, thirty-day cooldown, fourteen-day probe, three-to-five trial warm-up — are calibrations within the architecture, not the architecture itself.

Alternative inter-session intervals (e.g., fourteen-day minimum for additional buffer; ten-to-twenty-eight-day window for tighter standardisation) are compatible provided the lower bound is at least seven times the consolidation window. Alternative training-to-retest cooldowns (sixty days; twenty-one days for shorter operational cycles) are compatible provided the cooldown remains a substantial multiple of the consolidation window. Alternative retention-probe intervals (7, 14, 21, 30, 60, 90, 180 days) are explicitly enumerated as compatible variants.

Alternative fast-learning mitigation strategies are compatible: requiring multiple complete sessions before the first scored session with the first one or two designated as familiarisation; extending warm-up to a separate familiarisation session entirely; or statistical correction of first-session scores against a fast-learning model fitted to warm-up performance.

The architecture extends beyond colour-competency assessment to any platform integrating measurement with perceptual or cognitive training, including audiological assessment-and-training, motor-skill evaluation, sensory-substitution training, and skill-based professional certification. The Karni and Sagi (1993) findings concern texture-discrimination learning specifically; the method generalises to colour-discrimination learning under the conservative assumption that colour-discrimination consolidation is not substantially faster than texture-discrimination consolidation. For learning processes whose consolidation timescale is independently characterised, the rules can be calibrated to that process's specific timescale.

4. Cross-Innovation Architecture

4.1 Dependency graph

The three-layer competency framework (§3.1) is the architecture root. It defines what is measured; the remaining innovations address how each layer is measured and how results are reported. The 3×3 matrix (§3.2) depends on §3.1 directly and on §3.7 via per-instrument dimension mapping. The differentiated three-track validation strategy (§3.3) depends on §3.1 and is instrument-independent.

The Hue Ordering Diagnostic (§3.4) is a discrimination-layer instrument and depends on §3.7 for OKLCH cap construction. The two-path calibration architecture (§3.5) is layer-independent and is a precondition for valid measurement across all instruments. Cardinal-nearest hue classification (§3.6) scores attribution on the hue dimension and depends on §3.8 for cardinal-anchor values. OKLCH gamut-aware trial generation (§3.7) is a stimulus-construction primitive used by §3.4, §3.6, and the lightness/chroma ordering diagnostics. Kuehni-anchored attribution scoring (§3.8) provides the anchor values consumed by §3.6.

The temporal architecture (§3.9) operates orthogonally to the eight measurement-and-reporting innovations. It depends on §3.3 for the validation-track structure driving Track 1's two-session protocol but constrains all instruments uniformly through warm-up requirement and through inter-session, training-to-retest, and retention-probe intervals applied at platform level rather than instrument level. §3.9 is the only innovation addressing the time dimension of assessment; the others address what is measured (§3.1, §3.2), how validation accumulates (§3.3), specific instruments and stimulus generation (§3.4, §3.7), calibration basis (§3.5), and attribution-response scoring (§3.6, §3.8).

4.2 Composite architecture

Each of the nine innovations is individually implementable and individually useful. The composite — three-layer measurement, 3×3 matrix reporting, differentiated per-layer validation, two-path calibration, gamut-aware stimulus generation, cardinal-nearest classification, Kuehni-anchored scoring, and consolidation-grounded temporal architecture integrated in a single deployable platform — is a distinct subject of disclosure. No known published assessment architecture combines these elements in an integrated digital platform before this publication. The composite is placed in the public domain as prior art.

4.3 Relationship to the reference implementation

The ColorAptitude platform, developed by the author and operated through Nederlandse Kleurenschool and ColorExpertsHub, is the reference implementation. The distinction between disclosed architecture and retained implementation is as follows.

The architecture — methods, data structures, scoring formulas, reporting formats, calibration protocols, validation strategies, scheduling rules — is the subject of this defensive publication and is placed in the public domain as prior art. The implementation — source code, user-interface designs, proprietary datasets including normative data to be collected, server infrastructure, brand marks (ColorAptitude, Xconea, SNKI, Nederlandse Kleurenschool, ColorExpertsHub), and third-party integrations — remains the exclusive property of the author and Xconea.

Third parties may implement the disclosed architecture in their own source code, UI, and datasets under their own brand without infringing the architecture claims. They may not copy the ColorAptitude source code, reuse the brand marks, or access the proprietary datasets. Publication of the architecture prevents patent monopolisation; retention of the implementation preserves commercial value.

5. Legal Framework and Publication Record

5.1 Intellectual property ownership and management

The methods, architectures, and assessment frameworks disclosed are developed by Markus Kotterink (ORCID: 0009-0001-4225-5891). IP is owned personally; management, licensing, and commercial exploitation are conducted through Xconea — BNK Foundation, which holds an exclusive management mandate. Stichting Nederlands Kleur Instituut (SNKI) operates as a sublicensee of Xconea for specific commercial applications including training and certification through Nederlandse Kleurenschool and ColorExpertsHub. This three-tier structure predates this publication and is not asserted as a claim of this document.

5.2 Nature and purpose of this publication

This document is a defensive publication. Its purpose is establishment of public prior art for the subject matter disclosed in Sections 3.1 through 3.9. By publishing a dated, attributed, and enabling description of these innovations, the document places the described subject matter into the public domain with respect to novelty considerations in patent examination worldwide. Under 35 U.S.C. § 102(a)(1) and corresponding provisions in the European Patent Convention (Article 54), the EUTM Regulation, and comparable instruments in other jurisdictions, publicly available descriptions with an established publication date constitute prior art against subsequent patent claims.

5.3 Scope of disclosure versus scope of retained rights

The architecture disclosed in Sections 3.1 through 3.9 — methods, data structures, scoring formulas, reporting formats, calibration protocols, validation strategies, scheduling rules — is placed in the public domain as prior art. The implementation — source code, user-interface designs, proprietary datasets including normative data to be

collected, server infrastructure, brand marks, and third-party integrations — remains the exclusive property of the author and Xcone. Third parties may implement the disclosed architecture in their own systems without infringing the architecture claims. They may not copy the ColorAptitude source code, reuse Xcone/SNKI/Nederlandse Kleurenschool/ColorExpertsHub/ColorAptitude brand marks, or access the proprietary datasets.

5.4 Copyright and licensing

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5.6 Version and integrity

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Changelog v1.0 → v1.1: Addition of Section 3.9 (Temporal Architecture for Assessment, Training, and Re-Test) grounded in the consolidation kinetics of perceptual learning (Karni & Sagi, 1993), with corresponding cross-references in Sections 3.3, 4.1, 4.2, and the abstract. The eight innovations of v1.0 (Sections 3.1 through 3.8) and their disclosure dates as established by v1.0 publication are unchanged.

5.7 Citation

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5.8 Warranties and contact

This document is published without warranties of any kind. The author and Xcone make no representations about the validity, scope, or enforceability of any patent application, issued patent, or other intellectual property right, whether their own or of third parties, in connection with the subject matter disclosed. For questions regarding the

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